

Current Position

Postdoctoral Fellow,

Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA.

Supervisor(s): Aki Nishimura & Michael Rosenblum

Research collaboration,

University of California, Los Angeles, Los Angeles, USA.

Collaborator: Prof. Sudipto Banerjee

Research Interest

Bayesian Statistics, Spatial and Spatiotemporal Statistics, Bayesian Transfer Learning, Large-scale Statistical Modeling, Scalable Bayesian Computation, Statistical Learning.

Education

2021–2026 : **Ph.D. in Statistics,**

University of Milano–Bicocca, Milan, Italy.

Thesis: Bayesian Transfer Learning Approaches for Large-scale Spatiotemporal Problems

Supervisor(s) : Tommaso Rigon & Sudipto Banerjee

2019–2021 : **M.Sc. in Statistical & Economics Sciences,**

University of Milano–Bicocca, Milan, Italy.

Thesis: Models for spatial network data with measurement error in covariates: a Bayesian approach with INLA for road accidents

Supervisor(s) : Riccardo Borgoni & Andrea Gilardi

Grade : 110/110 cum laude

2016–2019 : **B.Sc. in Statistical Sciences,**

Alma Mater Studiorum - University of Bologna, Bologna, Italy.

Thesis: Structural models for the analysis of Italian employment rate

Supervisor(s) : Silvia Bianconcini

Grade : 110/110 cum laude

Publications

Journal Articles

2026 **Luca Presicce** and Sudipto Banerjee. *Bayesian Transfer Learning for Artificially Intelligent Geospatial Systems: A Predictive Stacking Approach.* **Journal of Machine Learning Research**, 2026. (to appear).

2023 Andrea Gilardi, Riccardo Borgoni, **Luca Presicce**, and Jorge Mateu. *Measurement error models for spatial network lattice data: Analysis of car crashes in Leeds.* **Journal of the Royal Statistical Society Series A: Statistics in Society**, volume 186, pages 313–334, 04 2023.

Submitted and Working papers

- 2026+ **Luca Presicce**, Andrew Chin, and Akihiko Nishimura. *Sticky Dynamical Monte Carlo for Structured Bayesian Variable Selection: with Application to Exploration of Gene-by-Environment Interactions*. (In preparation), 2026+.
- 2026+ **Luca Presicce** and Sudipto Banerjee. *Dynamic Bayesian Predictive Stacking via Markovian Spatiotemporal Propagation*. **arXiv preprint**. (under review), 2026+.
- 2026+ Lorenzo Tedesco and **Luca Presicce**. *Learning the Geometry of Nearest Neighbor Gaussian Processes*. (In preparation), 2026+.

Publications in monographs, volumes, and discussions

- 2025 **Luca Presicce** and Sudipto Banerjee. *Bayesian meta-learning approach for feasible large spatial analysis*, pages 442–448. **Methodological and Applied Statistics and Demography IV**, 2025.
- 2025 **Luca Presicce** and Sudipto Banerjee. *Variational Propagation for Exact Spatiotemporal Dynamic Modeling*. In Enrico di Bella, Vincenzo Gioia, Corrado Lagazio, and Susanna Zaccarin, editors, *Springer*, pages 61–67. **Statistics for Innovation IV**, 2025.
- 2025 **Luca Presicce**. *Enhancing Bayesian Distributed Learning with Spatially Adjusted Predictive Distributions*. In Enrico di Bella, Vincenzo Gioia, Corrado Lagazio, and Susanna Zaccarin, editors, *Springer*, pages 220–226. **Statistics for Innovation IV**, 2025.

National Conference Proceedings

- 2024 **Luca Presicce**, Tommaso Rigon, and Emanuele Aliverti. *Bias-reduction methods for Poisson regression models*. In **SIS 2023 book of the short papers**, pages 908–912. Pearson, 2024.
- 2022 Andrea Gilardi, Riccardo Borgoni, **Luca Presicce**, and Jorge Mateu. *Measurement error models on spatial network lattices: car crashes in Leeds*. In **CLADAG 2021 book of abstracts and short papers**, pages 348–351. Pearson, 2022.

Software and Repositories

- 2026 **Luca Presicce** and Sudipto Banerjee. *spFFBS: Spatiotemporal Propagation for Multivariate Bayesian Dynamic Learning*, 2026. **CRAN R** package version 1.0-1.
- 2025 **Luca Presicce** and Sudipto Banerjee. *spBPS: Bayesian Predictive Stacking for Scalable Geospatial Transfer Learning*, 2025. **CRAN R** package version 2.0-1.
- 2025 **Luca Presicce** and Sudipto Banerjee. *Markovian-Spatiotemporal-Propagation: Reproducibility Material of "Adaptive Markovian Spatiotemporal Transfer Learning in Multivariate Bayesian Modeling"*, 2025. Github repository.
- 2024 **Luca Presicce** and Sudipto Banerjee. *Bayesian-Transfer-Learning-for-GeoAI: Reproducibility Material of "Bayesian Transfer Learning for Artificially Intelligent Geospatial Systems: A Predictive Stacking Approach"*, 2024. Github repository.

Awards

Data Competition

June 2023 : **Best Objective Prediction**, *Stats under the Stars 6*, Ancona, Italy.

Travel Awards

March 2023 : **Travel award**, *Bayes Comp*, Levi, Finland.

July 2024 : **Travel award**, *ISBA World meeting*, Venice, Italy.

July 2026 : **Travel award**, *ISBA World meeting*, Nagoya, Japan.

Funding and Grants

Team member of the research group, in the *Milano–Bicocca* local unit of the PRIN-PNRR grant "*Measuring Biodiversity via Bayesian Nonparametrics: Estimation, Clustering and Uncertainty Quantification*" funded by MIUR., Local PI: Federico Camerlenghi, PI: Igor Prünster.

Presentations

Invited Seminars

May 2025: **Karlsruhe Institute of Technology**, *Karlsruhe*, Germany.

February 2026: **John Hopkins University**, *Online*.

Invited Presentations

June 2025: **SIS 2025**, *Genova*, Italy.

July 2026: **ISBA 2026**, *Nagoya*, Japan.

May 2027: **Bayes Comp 2027**, *Texas*, USA.

Contributed Presentations

June 2023: **SIS 2023**, *Ancona*, Italy.

June 2024: **SIS 2024**, *Bari*, Italy.

April 2025: **BAYSM 2025**, *Online*.

June 2025: **SIS 2025**, *Genova*, Italy.

Poster Presentations

June 2022: **ISBA World meeting**, *Montréal*, Canada.

March 2023: **Bayes Comp**, *Levi*, Finland.

June 2023: **SIS 2023**, *Ancona*, Italy.

June 2024: **BAYSM 2024**, *Venice*, Italy.

July 2024: **ISBA World meeting**, *Venice*, Italy.

May 2025: **BISP**, *Milan*, Italy.

Sept 2025: **SISBayes**, *Padua*, Italy.

Sept 2025: **GRASPA**, *Rome*, Italy.

Service to Profession

Referee service

Annals of Applied Statistics (AoAS) · **Spatial statistics (SPASTA)**.

Position in Academic Societies

2019–2021 **Students representative of the MSc in Statistical & Economics Sciences (LM-82)**, *Department of Economics, Management and Statistics*, University of Milano–Bicocca.

Membership

2021 – present **International Society for Bayesian Analysis (ISBA)** · **The junior section of the International Society for Bayesian Analysis (j-ISBA)** · **Institute of Mathematical Statistics** · **Società Italiana Statistica (SIS)** · **young SIS (y-SIS)** .

Teaching Experience

As Teaching Assistant

November 2021 – October 2023: **Statistics II - Tutor & Lecture activities**, *B.Sc. in Statistical & Economics Sciences*, University of Milano–Bicocca.

- January 2025 – **Statistical Learning - Tutor & Lecture activities**, *M.Sc. in Data analytics for business*, Catholic University of Sacred Heart.
September 2025:
- September 2025 – **Statistics I - Tutor & Lecture activities**, *B.Sc. in Economics*, Catholic University of Sacred Heart.
– February 2026:
- [As Lecturer](#)
- September 2021 – **PLS - Scientific Degree Plan - Lectures & Laboratory**, *High schools*, University of Milano–Bicocca.
– September 2025:
- January 2023 – **PNRR - Orientation Program - Lectures & Laboratory**, *High schools*, University of Milano–Bicocca.
September 2025:

Working Experience

- May – August 2021: **Data Scientist**, *Elmec Informatica Spa*, Research in AI for IT security , Brunello (VA).
- July – July 2019: **Data Analyst**, *Rapelli SA*, Analysis on production, sales and quality control, Stabio (CH).
- June – August 2018: **Corporate Analyst**, *REMAX Altoverbano*, Business analysis, Germignaga (VA).

Summer Schools and Workshops

Schools

- 2022 **Lake Como school of advanced studies**,
RSFD - Robust Statistics: Foundations and recent Developments, Como, May 2–6, 2022.

Workshops

- 2023 **Workshop at Collegio carlo Alberto**,
Club-PMx² - CLustering: Bayesian Partition Models for Precise Medicine, Torino, February 24, 2023.
- 2023 **Workshop at Bocconi University**,
Theory for Scalable, Modern Statistical Methods, Milano, April 5–7, 2023.
- 2024 **Workshop at USI**,
Satellite workshop to ISBA World meeting, Lugano, June 25–28, 2024.